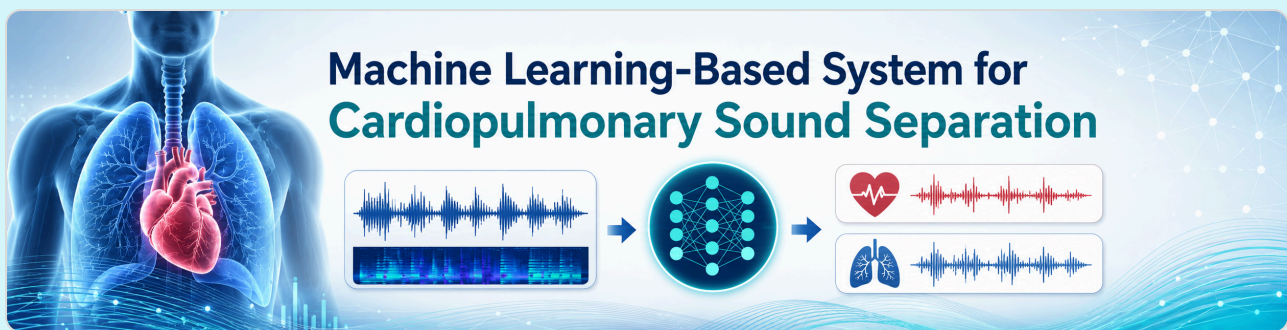




User Requirements Questionnaire for Machine Learning-Based Cardiopulmonary Sound Separation System

53 responses

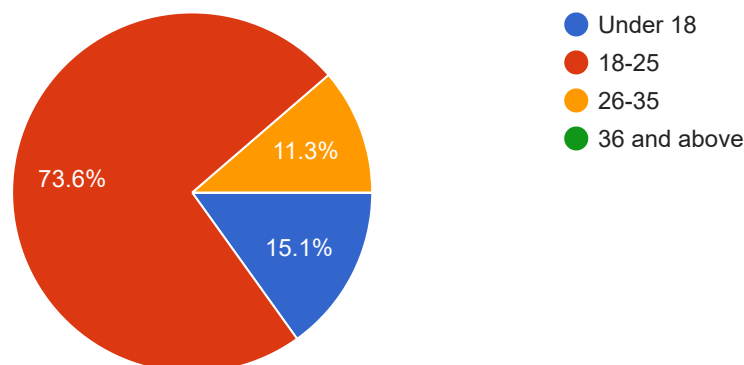
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Section A - Respondent Background:

1. What is your age group?

[Copy chart](#)

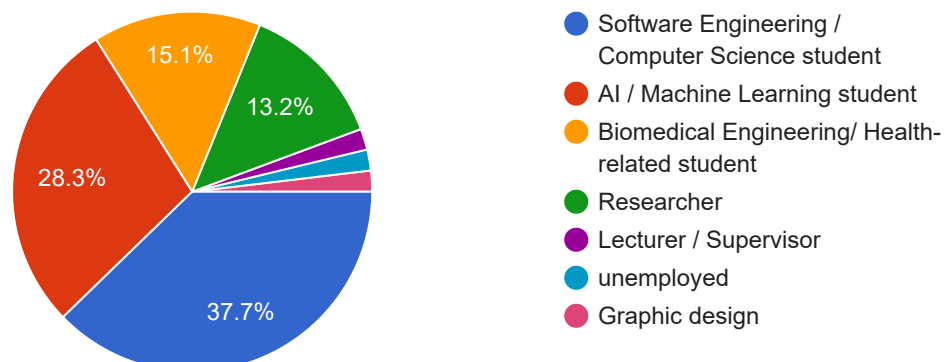
53 responses



2. What best describes you?

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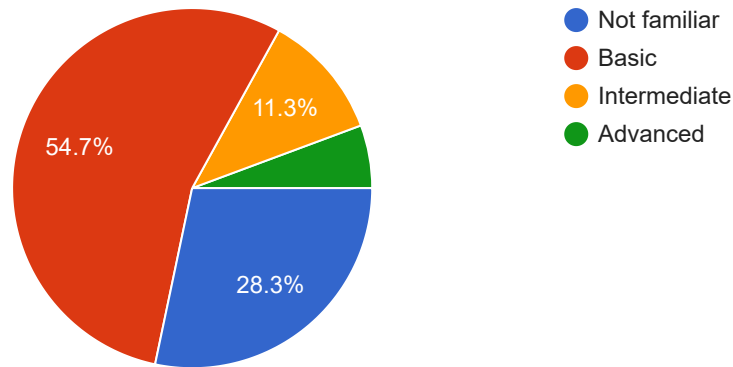
53 responses



3. How familiar are you with audio processing or signal processing?

 [Copy chart](#)

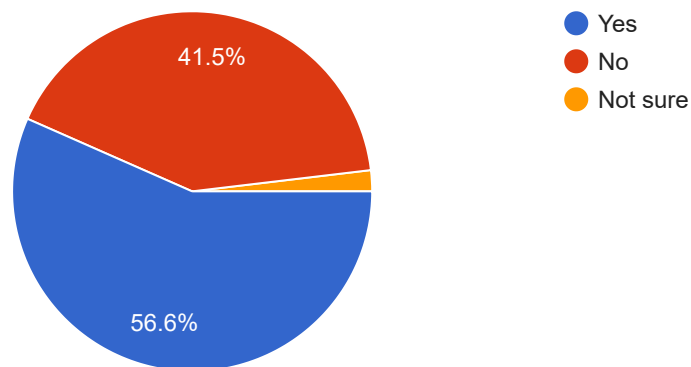
53 responses



4. Have you worked with biomedical audio before?

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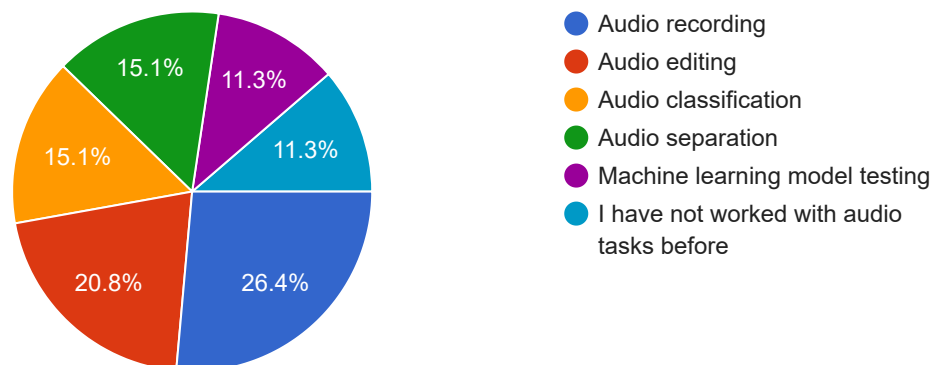
53 responses



5. Which type of audio-related task are you most familiar with?

 [Copy chart](#)

53 responses



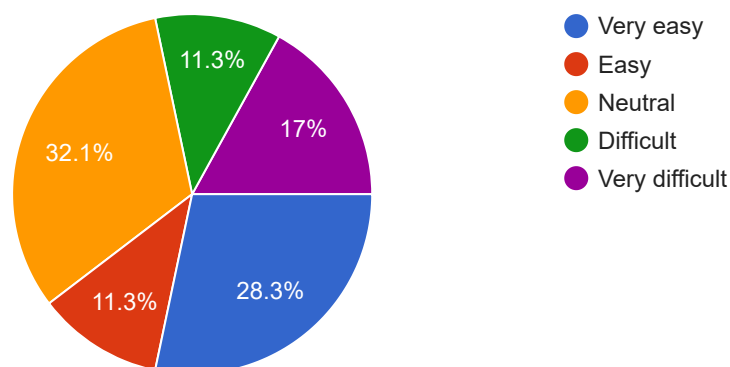
Section B - Current Challenges in Cardiopulmonary AudioProcessing:



1. How difficult do you think it is to separate heart and lung sounds from a mixed cardiopulmonary recording?

 [Copy chart](#)

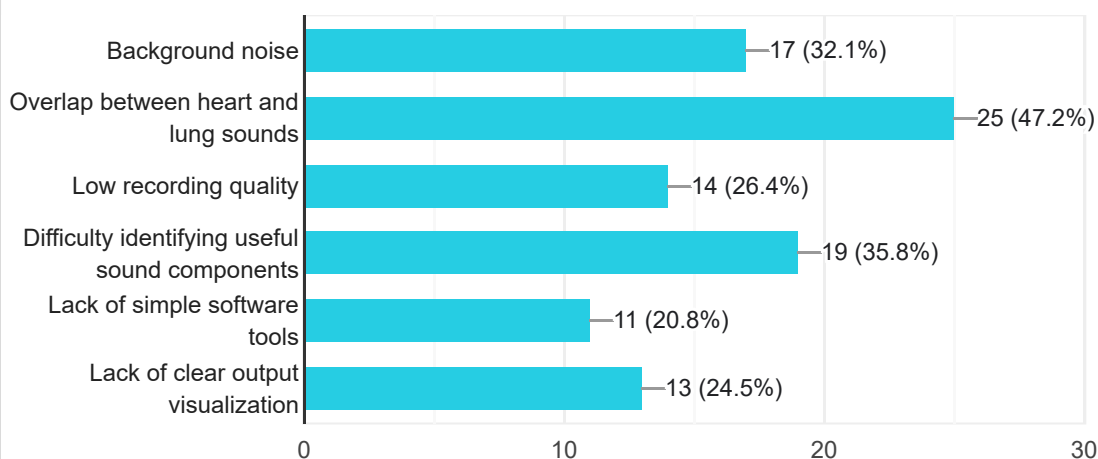
53 responses



2. What problems do you think affect cardiopulmonary sound processing the most?

 [Copy chart](#)

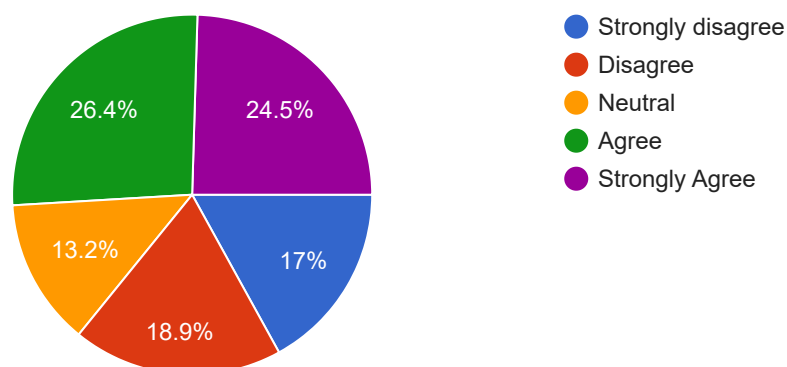
53 responses



3. Do you think noisy audio recordings can reduce the quality of machine learning-based sound processing?

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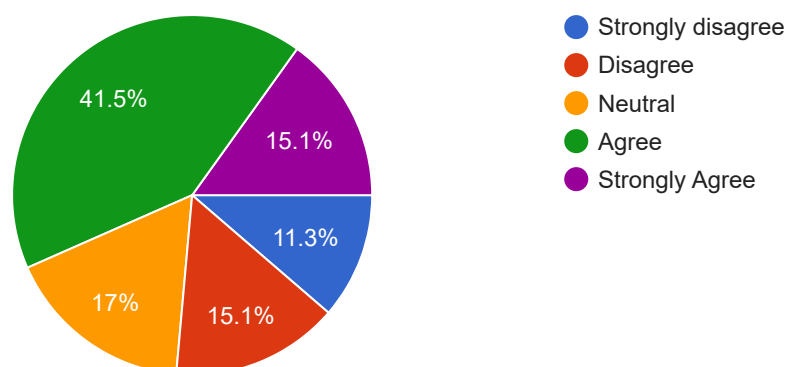
53 responses



4. Do you think a simple software prototype for heart-lung sound separation would be useful for students or researchers?

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53 responses

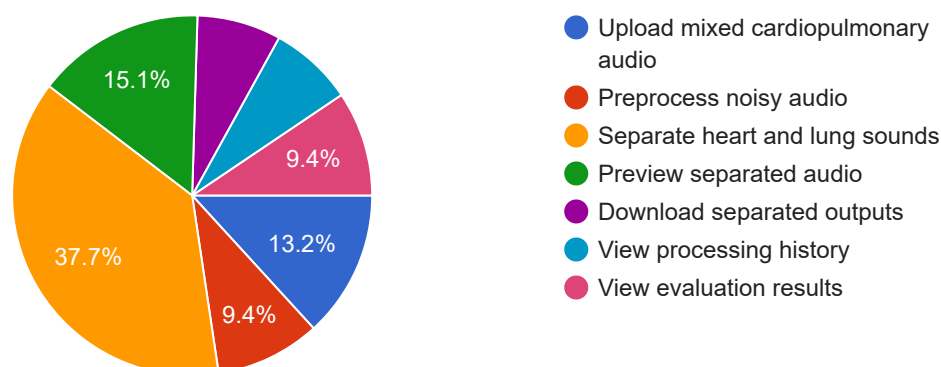


Section C - System Feature Preferences:

1. Which feature would be most important in the proposed system?

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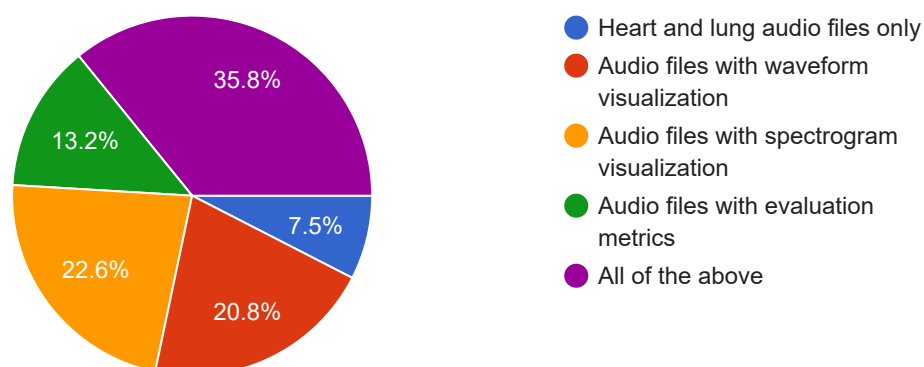
53 responses



2. Which output would you prefer after sound separation?

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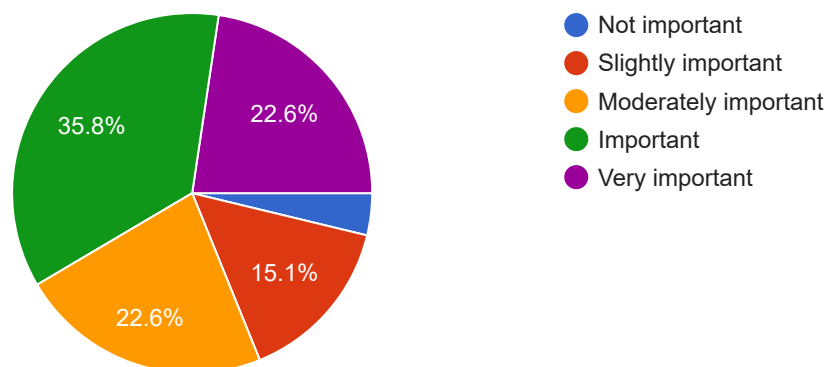
53 responses



3. How important is it for the system to provide a preview of the separated heart and lung sounds?

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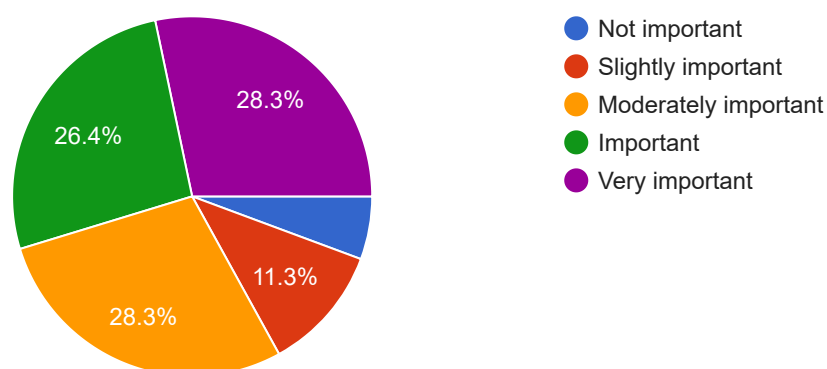
53 responses



4. How important is it for the system to allow users to download separated audio outputs?

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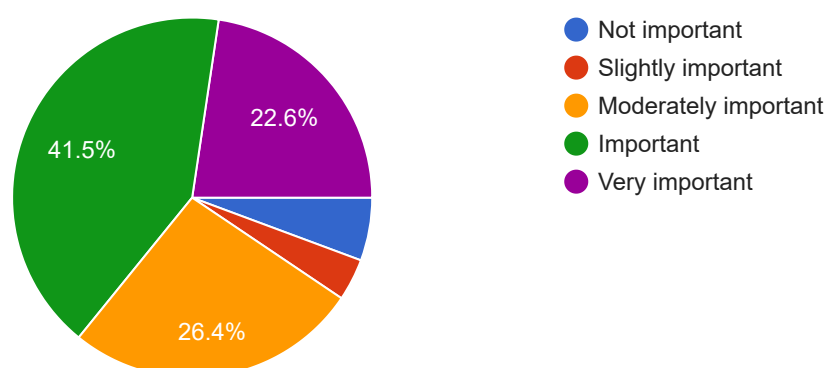
53 responses



5. How important is it for the system to show basic processing information, such as file name, processing status, and completion time?

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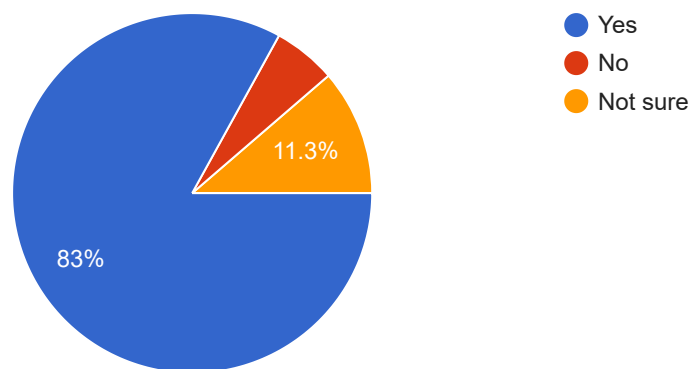
53 responses



6. Should the system include a history page showing previously uploaded files and separation results?

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53 responses

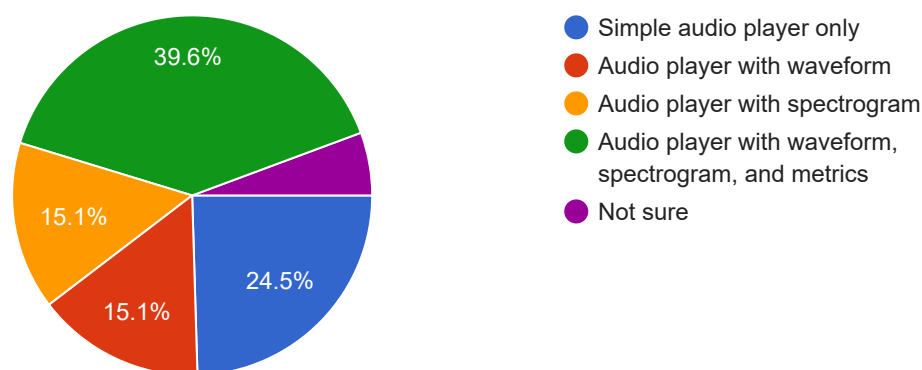


Section D - Interface and Output Preferences:

1. How would you prefer the separated results to be displayed?

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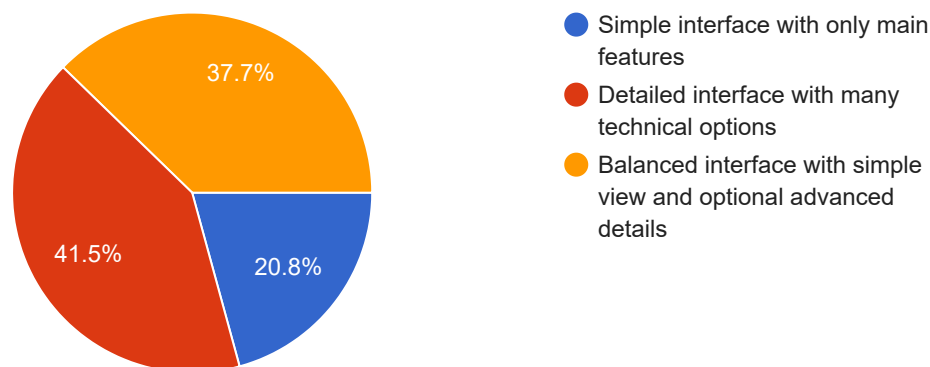
53 responses



2. Which interface style would you prefer?

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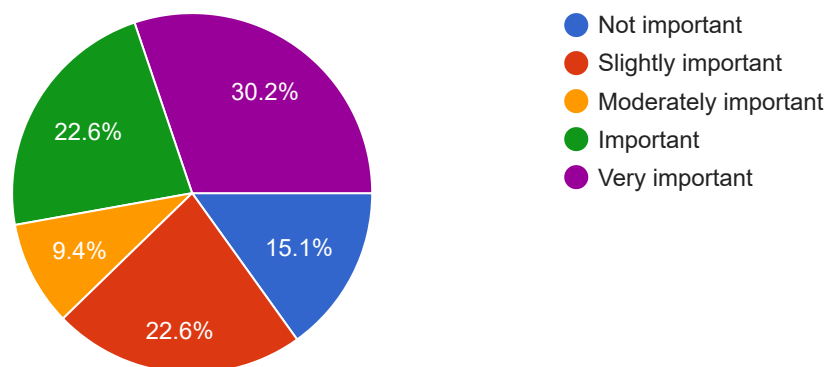
53 responses



3. How important is it for the system interface to be easy for non-expert users?

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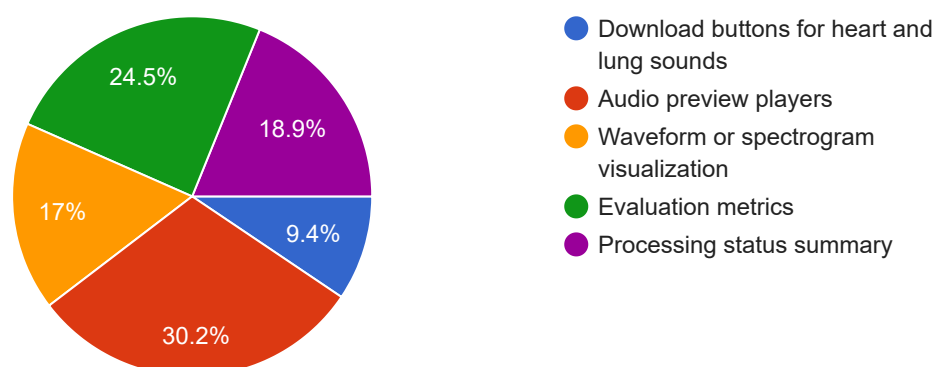
53 responses



4. What information should appear first after separation is completed?

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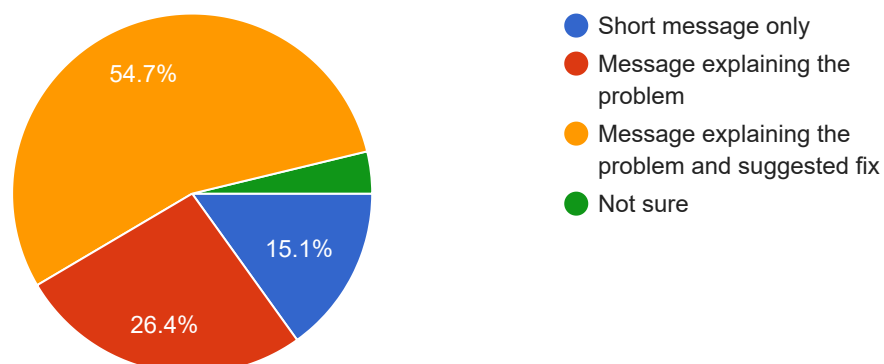
53 responses



5. What type of error message would be most helpful if the uploaded file is invalid?

 [Copy chart](#)

53 responses



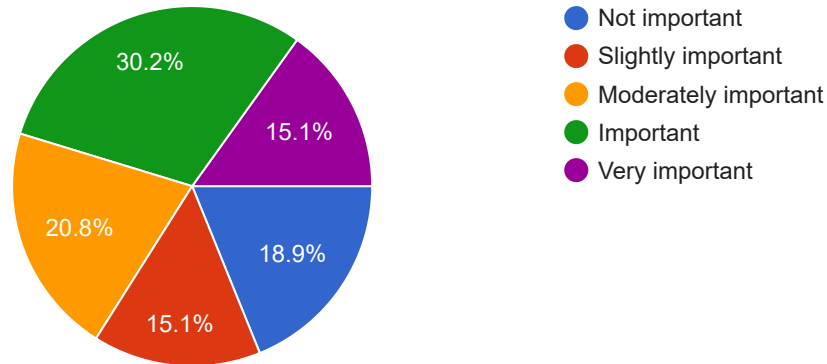
Section E - Evaluation and Improvement:



1. How important is it for the system to show evaluation metrics for the separated output?

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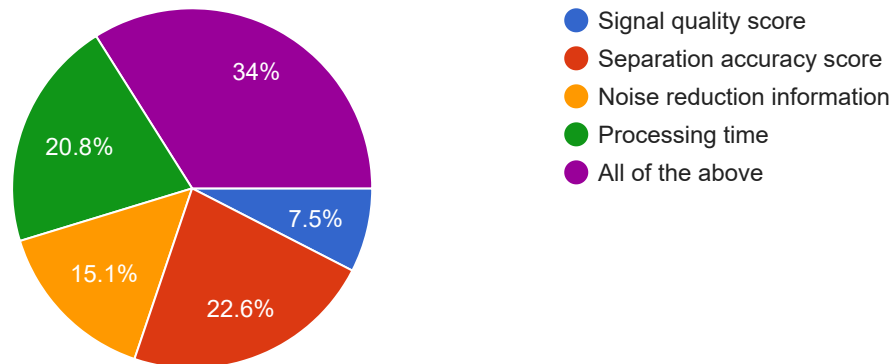
53 responses



2. Which evaluation information would be most useful?

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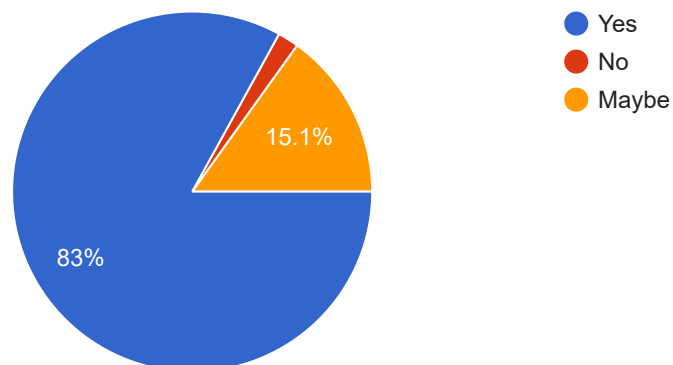
53 responses



3. Would you use a prototype that separates mixed cardiopulmonary audio into heart and lung sound outputs?

 [Copy chart](#)

53 responses



4. What improvement or feature would you like to see in a cardiopulmonary sound separation system?

18 responses

Anything that can help the human being

It's ok

Not sure

You 😊

I would like it to be easier for non-technical users to be able to understand it and use it with no need for technical or advanced knowledge.

idk

everything 😊

I would like to see the integration of real-time, adaptive AI noise cancellation that can perfectly isolate heart and lung sounds even in noisy, chaotic clinical environments (like emergency rooms or ambulances). Additionally, embedding an automated anomaly detection feature that not only separates the sounds but also highlights potential murmurs or wheezing instantly would be a game-changer for fast diagnostics

An important feature would be noise reduction and improved accuracy of separation between heart and lung sounds, making diagnosis clearer and more reliable

Better preservation of subtle heart murmurs and faint lung crackles.
Minimal artifacts introduced during separation.

-

I would like the system to include clear waveform and spectrogram visualization, audio preview, download options, evaluation metrics, and simple explanations of the separation results so that both students and researchers can understand the output easily.

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